Face, Content, and Construct Validity of a Novel Portable Ergonomic Simulator for Basic Laparoscopic Skills

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OBJECTIVE: Laparoscopic skills can be improved effectively through laparoscopic simulation. The purpose of this study was to verify the face and content validity of a new portable Ergonomic Laparoscopic Skills simulator (Ergo-Lap simulator) and assess the construct validity of the Ergo-Lap simulator in 4 basic skills tasks.

DESIGN: Four tasks were evaluated: 2 different translocation exercises (a basic bimanual exercise and a challenging single-handed exercise), an exercise involving tissue manipulation under tension, and a needle-handling exercise. Task performance was analyzed according to speed and accuracy. The participants rated the usability and didactic value of each task and the Ergo-Lap simulator along a 5-point Likert scale.

SETTING: Institutional academic medical center with its affiliated general surgery residency.

PARTICIPANTS: Forty-six participants were allotted into 2 groups: a Novice group ($n = 26$, <10 clinical laparoscopic procedures) and an Experienced group ($n = 20$, >50 clinical laparoscopic procedures).

RESULTS: The Experienced group completed all tasks in less time than the Novice group did ($p < 0.001$, Mann-Whitney U test). The Experienced group also completed tasks 1, 2, and 4 with fewer errors than the Novice group did ($p < 0.05$). Of the Novice participants, 96% considered that the present Ergo-Lap simulator could encourage more frequent practice of laparoscopic skills. In addition, 92% would like to purchase this simulator. All of the experienced participants confirmed that the Ergo-Lap simulator was easy to use and useful for practicing basic laparoscopic skills in an ergonomic manner. Most (95%) of these respondents would recommend this simulator to other surgical trainees.

CONCLUSIONS: This Ergo-Lap simulator with multiple tasks was rated as a useful training tool that can distinguish between various levels of laparoscopic expertise. The Ergo-Lap simulator is also an inexpensive alternative, which surgical trainees could use to update their skills in the skills laboratory, at home, or in the office. (J Surg 71:65-72. © 2014 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights reserved.)

KEY WORDS: ergonomic, laparoscopy, simulator, surgical trainee

COMPETENCIES: Medical Knowledge, Practice-Based Learning and Improvement, Interpersonal and Communication Skills

INTRODUCTION

Technique and skills are essential to be able to perform laparoscopic procedures safely and effectively. One practical solution for acquiring these skills involves the use of a laparoscopic simulator.[1] Studies have shown that specialists can improve their clinical performance through dedicated practice of their laparoscopic skills.[2]

Several types of simulators are currently being used for training purposes: traditional box trainers, virtual reality (VR), and augmented reality simulators. These simulators play important roles in surgical education.[3,4] Several useful yet expensive VR and augmented reality simulators (e.g.,

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Source of funding: PhD project of Delft University of Technology.

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Surgical Science AB LapSim, Simbionix LAP Mentor, and CAE Healthcare ProMIS) are commercially available, along with training modules ranging from basic skills to complex laparoscopic procedures. With prices ranging from €1560 to €70,000,[5,6] however, these tools are too expensive for many hospitals, particularly in developing countries.

Many medical centers also use inexpensive box trainers for practicing basic and advanced laparoscopic skills, although these tools are somewhat limited in their application. For example, the Karl Storz box trainer with multiple tasks, which has been used in the skills laboratory of Catharina Hospital Eindhoven, requires a special display, cables, or other cumbersome equipment. Various efforts have been devoted to developing portable box simulators that can improve the convenience and efficiency of surgical training. Examples include the portable Fundamentals of Laparoscopic Surgery box trainer developed by the Fundamentals of Laparoscopic Surgery program,[7] the mirrored-box simulator produced by Keyser et al.,[8] and the portable trainer assessed by Hruby et al.[9] Several investigations have focused on homemade simulators involving cards or plastic box trainers.[1,10-12] These portable box trainers provide convenient and effective alternatives with which trainees can practice basic skills at home or in the office.

One major shortcoming of the simulators that are currently available is that they do not support the ergonomic guidelines for laparoscopic surgery, thus failing to offer an ergonomic training setup.[13,14] According to the studies by Hanna et al., the performance of bimanual tasks could be improved by obtaining an optimal manipulation angle of 60° (with equal azimuth angles) and an optimal elevation angle between 30° and 60°.[1,15] Furthermore, according to a study by Emma et al.,[16] intracorporal/extracorporal instrument length ratios below 1.0 cause deterioration in task performance, expand the range of movement at the elbow and shoulder, and increase angular velocity at the shoulder. It is important for trainees to become accustomed to proper ergonomic working conditions, as this can improve task performance and minimize physical discomfort.[14]

Nevertheless, a portable and ergonomic simulator has yet to become available. The authors have developed a new portable Ergonomic Laparoscopic Skills simulator (Ergo-Lap simulator) from a user-centered perspective, following the ergonomic guidelines.[13,16]

Before a surgical simulator can be used to assess surgical skills, its validity must be subjected to rigorous verification.[17,18] This study focuses on 3 basic and valuable forms of validity: construct validity, face validity, and content validity. Construct validity indicates the extent to which the simulator can distinguish between different levels of expertise based on the performance scores. Content validity indicates the extent to which the simulator is able to teach the trainees what it is intended to teach, as evaluated by experts dedicated to the device. Face validity refers to the relative realism of the simulator, as evaluated by nonexperts.[18] This study aims to verify the construct validity of 4 innovative tasks on the Ergo-Lap simulator for training basic laparoscopic skills, in addition to evaluating the face validity and content validity of this new ergonomic portable simulator.

MATERIALS AND METHODS

Ergo-Lap Simulator

In the context of this study, the Ergo-Lap simulator with multiple tasks is used to train basic laparoscopic skills. Its dimensions are similar to those of a folded briefcase (55 cm × 32 cm × 12 cm), and it can be stored in limited space (Fig. 1). The system weighs 3 kg, including the task materials and an integrated handle for easy transportation. The simulator can be linked to a computer, where video

FIGURE 1. Ergo-Lap simulator: The Ergo-Lap simulator ready for use (left); the flexible arrangement of the task panel allows for ergonomic training conditions (upper right); the Ergo-Lap simulator ready for transportation (bottom right).

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software makes it easy to record task performance, thus allowing surgical trainees to evaluate and improve their performance. The physical setup of the Ergo-Lap simulator can be adjusted to attain an ergonomic setting and adapt to various types of training tasks for a broad range of users (Fig. 1). For example, the task panel can move backward/forward, and it can be tilted at various angles to acquire a proper optical axis-to-target viewing angle and an appropriate intracorporal/extracorporal instrument length ratio. The cover height of the simulator can also be adjusted to create different volumes of the workspace. The simulator is accompanied by an instruction manual, which helps trainees to create an optimal ergonomic training setup for each task.

Tasks and Assessment

Four tasks were selected for validation in this study (Fig. 2). Task 1 (transfer of balls) was designed to involve the use of graspers and the development of bimanual coordination skills. Task 2 (transfer of tubes) was designed to involve the use of graspers by the dominant and nondominant hands and to train surgeons to adapt their perceptions from 3 dimensions to 2 dimensions. Task 3 (band stretch) was designed to develop laparoscopic coordination skills with both hands. The goal of Task 4 (pass suturing needle and a small ball through rings) was designed to involve the use of needle holders and to develop bimanual coordination skills. In tasks 1 and 2, any time a ball or tube is dropped, it is counted as an error. In task 3, any time the band slips or is lost, it is counted as an error. In task 4, errors are defined as dropping the needle, missing the rings, and passing the needle in the wrong direction. The completion time and error frequency of each task were recorded on video. This video was then used to assess task performance in terms of the completion time and number of errors. To avoid bias according to level of expertise, the observer was blinded to the skill level of the participant performing the tasks.

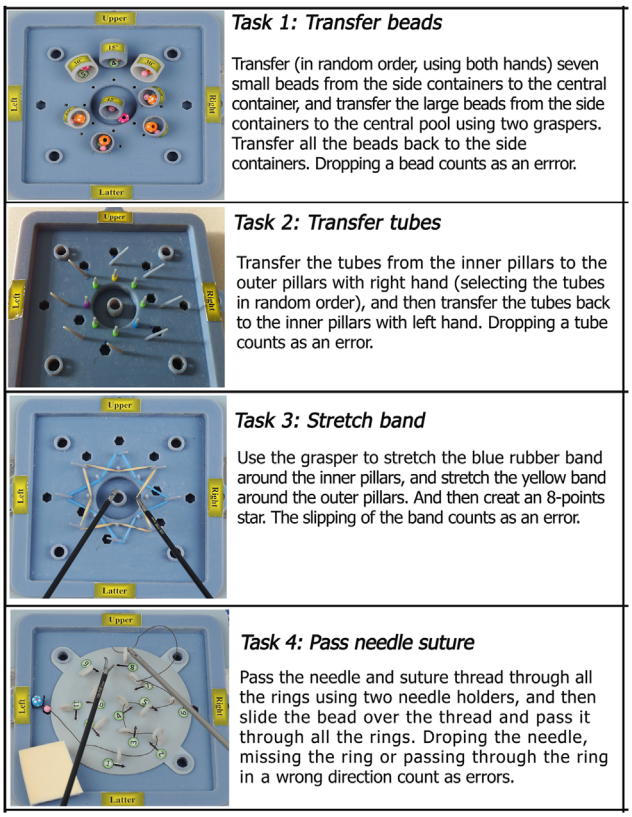

FIGURE 2. Overview of the 4 Ergo-Lap simulator exercises used in this study.

Subjects

The Ergo-Lap simulator was brought to 2 hospitals: the Catharina Hospital Eindhoven and the Máxima Medical Centre, both in the Netherlands. Junior and senior residents, medical interns, and laparoscopic specialists were invited to perform several tasks on the Ergo-Lap simulator during the intervals of their workday. The participants were divided into 2 groups according to their clinical laparoscopic experience: a Novice group, consisting of participants who had performed fewer than 10 clinical laparoscopic procedures, and an Experienced group, comprising participants with clinical experience who have performed more than 50 laparoscopic procedures. For the basic skills tests, surgeons who had performed more than 50 procedures were assigned to the Experienced group.[19,20]

Protocol

All participants provided informed consent after having received an explanation of the study design. They completed questionnaires based on demographics and their previous laparoscopic experience. Before performing a task on the simulator, the participants received a general introduction to the Ergo-Lap simulator, as well as standardized instruction, including a video demonstration of each task. Nevertheless, some of the trainees did not have enough time to complete all 4 of the tasks. This was particularly the case for the experienced participants, who sometimes had to leave to treat emergency patients. For this reason, the sequence of the 4 tasks was randomized for each participant, thereby ensuring that each task was performed by at least 18 participants from each group. The minimum number of participants needed in each group (17) was determined according to a 2-tailed test ($\alpha = 0.05$; power $[1\text{-}\beta] = 0.80$), with a fixed effect size of 0.9 (following the study by Livingston and Cassidy).[20,21] This number was increased to 18 to compensate for potential attrition.

Figure 3 illustrates the process of this study. The participants performed each task twice, with only the score

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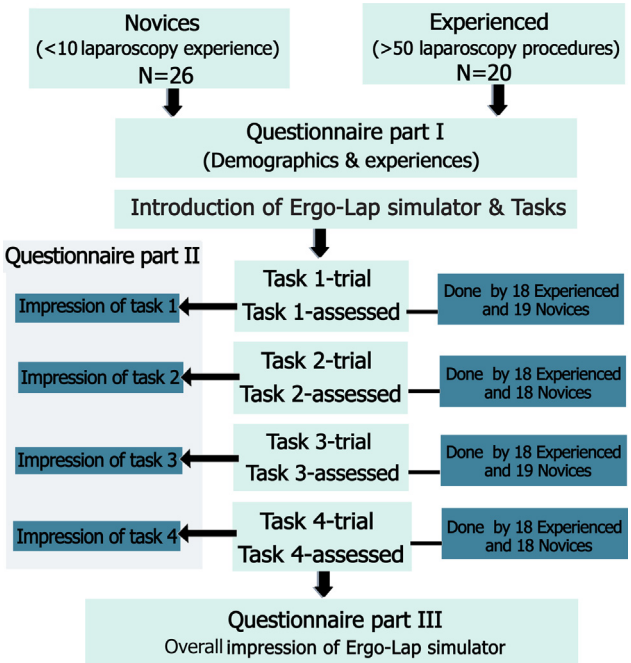

FIGURE 3. Overview of the study protocol.

from the second repetition used to evaluate the performance. For tasks 1, 2, and 3, participants used 2 graspers (type Karl Storz 33310), and for task 4, they used 2 needle holders (type Karl Storz 26173). After finishing each task, the participants completed the questionnaire and evaluated their task performance along a 5-point Likert scale (Table 1). After performing all of the tasks, the participants completed the last section of the questionnaire, which concerned their overall impressions of the Ergo-Lap simulator.

In addition, one item was included in the questionnaire to investigate what the participants would consider a reasonable price for the Ergo-Lap simulator.

Statistical Analysis

The completion time and the total number of errors were recorded on video for each task. Comparisons between the Novice and Experienced groups were performed using the software package SPSS, version 17.0 (SPSS, Chicago, IL). For normally distributed data, the Student *t* test was chosen; for all other data, the nonparametric Mann-Whitney U test was used. The significance level was set at $p < 0.05$.

RESULTS

In all, 46 participants took part in this study: 26 participants were assigned to the Novice group, which ultimately consisted of 14 medical interns and 8 junior surgical residents who had not yet performed any clinical procedures as well as 4 residents who had performed fewer than 10 clinical procedures. The Experienced group consisted of 20 participants, including 12 surgeons and 8 senior residents, each of whom had performed more than 50 clinical procedures.

Time and Error Score

Construct validity was evaluated by comparing the completion times and error frequencies of the Novice and Experienced groups. The normality test was used to analyze each group of data to determine whether the Student *t* test or the Mann-Whitney U test should be used. As shown in Figure 4, the Experienced group completed all of the tasks in significantly less time than the Novice group did (Student *t* test for tasks 1 and 4; Mann-Whitney U test for tasks 2 and 3), and they performed tasks 1, 2, and 4 with significantly fewer errors (Mann-Whitney U test for tasks 1 and 4; Student *t* test for task 2), as shown in Figure 5. The results presented in Figures. 4 and 5 illustrate that the Experienced group's completion time and error frequency were centralized with a narrow standard deviation (SD),

TABLE 1. Participants' Overall Evaluations of Each Task, Rated Along a 5-Point Likert Scale (1 = Strongly Disagree, 3 = Neutral, 5 = Strongly Agree). All values in mean (SD)

	Task 1	Task 2	Task 3	Task 4
The task is useful for working with graspers (for tasks 1, 2, and 3) and needle holders (for task 4).	4.14 (0.10)	4.03 (0.11)	3.89 (0.13)	4.33 (0.11)
The task makes the trainee think about which hand to use (for task 1)	4.19 (0.11)			
The task is useful for training the ability to shift perception from 3D to 2D (for task 2).		4.22 (0.10)		
The task is useful for training hand coordination (for tasks 3 and 4).			4.13 (0.11)	4.17 (0.12)
How difficult do you consider this task?*	2.92 (0.16)	3.83 (0.14)	3.50 (0.13)	3.92 (0.16)
Global impression of the task†	4.08 (0.10)	3.97 (0.10)	4.00 (0.11)	4.11 (0.09)

3D, 3 dimension.
Results printed in boldface indicate significant differences.
*1 = very easy, 3 = neutral, and 5 = very difficult.
†1 = very poor, 3 = neutral, and 5 = excellent.

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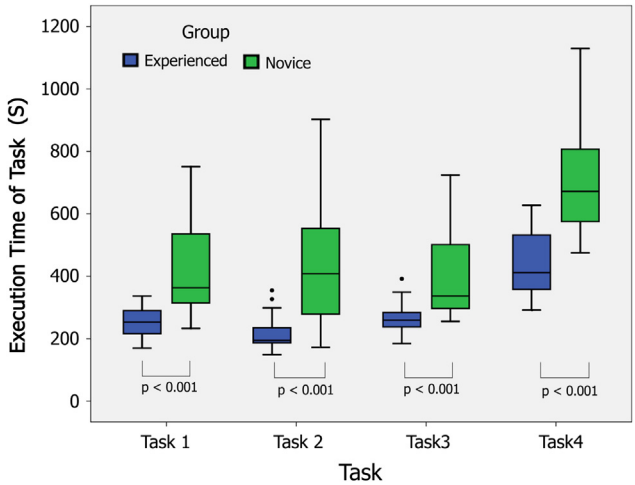

FIGURE 4. Completion time for each task and expertise group. Presented p values represent significant differences between the 2 expertise groups for each task (Mann-Whitney U test).

while the results from the Novice group were more scattered.

Participant Opinions Regarding the Tasks

All of the participants rated their global impressions of all 4 tasks as good to excellent (Table 1). Insignificant differences were found between the ratings of the Experienced group and those of the Novice group, with the exception of the item concerning the level of difficulty. The Experienced group rated task 2 as more difficult than the Novice group did (means of 4.11 and 3.56, respectively; SD of 0.83 and 0.79, respectively; $p < 0.050$). The Novice group rated task 4 as more difficult than the Experienced group did (means of 4.44 and 3.39, respectively; SD of 0.51 and 1.04, respectively; $p < 0.050$). Tasks 1 and 3 were rated equally difficult by both groups.

Face and Content Validity of the Ergo-Lap Simulator

The face validity of the Ergo-Lap simulator was evaluated by the participants in the Novice group, with 92% rating the simulator's attractiveness as neutral to excellent. In addition, 96% of the Novice participants thought that the present Ergo-Lap simulator could lead to more frequent practice of laparoscopic skills. Furthermore, 92% reported that they would like to purchase the simulator (Table 2).

The Experienced group was asked several questions to assess the content validity of the Ergo-Lap simulator (Table 3). All of these participants confirmed that the Ergo-Lap simulator was easy to use and useful for practicing basic laparoscopic skills in an ergonomic fashion. Most (95%) of these participants said that they would recommend this simulator to other surgical trainees.

We also investigated the price that the participants would consider reasonable for the Ergo-Lap simulator. Of the 46 participants, 23 indicated a range of €400 to €600, while 13 indicated a range of €200 to €400. The anticipated price for the Ergo-Lap simulator is less than €300, based on the cost of the prototype.

DISCUSSION

Given the performance of the Experienced group was significantly better than was that of the Novice group, with shorter completion times and fewer errors for all 4 tasks, we can conclude that these tasks are capable of distinguishing between different levels of expertise.

The participants reported that tasks 1, 2, and 3 were very useful for training bimanual coordination. The video revealed that, in task 1, the Experienced group was better aware of which hand should be used to pick up the beads, as the surgical trainees were forced to decide which hand was appropriate for different targets.

In task 2, the video showed that the Novice group spend much more time aligning the orientation between the tube and the pillar. Participant opinions indicate that this task is very useful for training the surgeon's ability to adapt their perceptions from 3-dimensional to 2-dimensional.

The video further showed that, in task 3, the Novice group spent much more time coordinating 2 graspers to stretch the rubber bands than the Experienced group did. Task 4 involved a total of 12 rings with various orientations. The participants considered this task useful for

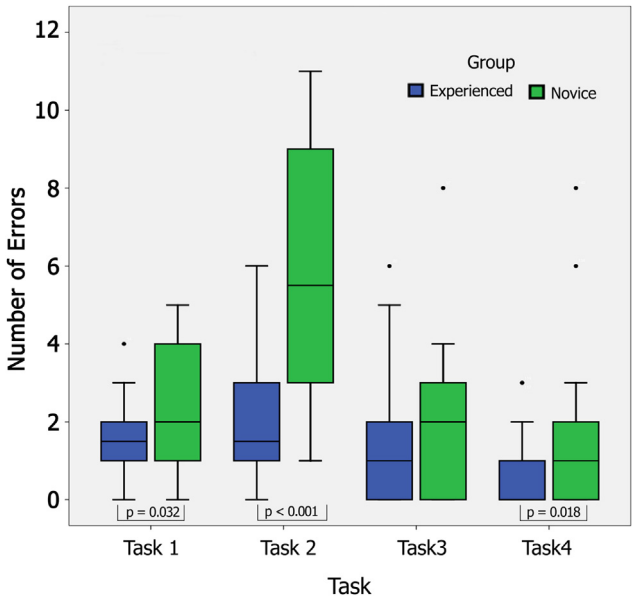

FIGURE 5. Number of errors for each task and expertise group. Presented p values represent significant differences between the 2 expertise groups for each task (Mann-Whitney U test and Student t test; only significant differences are presented).

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TABLE 2. Face Validity of the Ergo-Lap Simulator, as Assessed by the Novice Group ($n = 26$), Rated Along a 5-Point Likert Scale (1 = Strongly Disagree [Very Poor], 3 = Neutral, and 5 = Strongly Agree [Excellent])

	Responses				
	1	2	3	4	5
How would you rate the attractiveness of this Ergo-Lap simulator? *n* (%)	0	2 (8)	1 (4)	18 (69)	5 (19)
Do you think that having this Ergo-Lap simulator would encourage you to practice your laparoscopic skills more often? *n* (%)	0	1 (4)	5 (19)	13 (50)	7 (27)
Would you like to have an Ergo-Lap simulator of your own? *n* (%)	1 (4)	1 (4)	2 (8)	13 (50)	5 (34)

training surgical trainees to pass and hold the needle properly. In addition, the task using a small ball placed on a suture was also useful for training bimanual coordination. Compared with the tasks designed for other box simulators, the tasks designed for this assessment of the Ergo-Lab simulator were more challenging, and they were more effective in stimulating the trainees to improve and evaluate their performance.

The purpose of the current study was to evaluate the face validity, content validity, and construct validity of the Ergo-Lap simulator system, which includes 13 laparoscopic exercises from basic to advanced tasks. Of the 13 tasks included in the system, the 4 most innovative were chosen for the first step of the validation process, as they had been designed for basic psychomotor skills. Several researchers have validated laparoscopic simulators that can distinguish the performance scores of different expertise levels, in most cases basing their scores on completion time.[22,23] The use of completion time alone, however, is not sufficient for providing a precise measure of task performance. Other studies have assessed laparoscopic skills in terms of accuracy and speed, although they do not mention the reasoning underlying their calculations.[17,24-26] Although completion time and number of errors are both essential components of an objective assessment of task performance, it is difficult to determine the relative weights for these 2 indices when constituting the final scores. In this study, the task performance of the participants was graded according to completion time and number of errors.

Various studies have examined the process of learning surgical skills and the optimization of training. Stefanidis[27] reported on an investigation of the theoretical framework underlying the acquisition of surgical skills and factors that optimize learning on simulators. Brinkman et al.[28] demonstrated that the training of basic laparoscopic skills using single or multiple modalities does not produce different training outcomes. It should therefore be possible to acquire basic laparoscopic skills with simple box or home simulators.

The face validity of the Ergo-Lap simulator was evaluated by the Novice group, 96% of whom considered that the present Ergo-Lap simulator could increase the frequency of practicing laparoscopic skills. The Expert group rated the content validity of the Ergo-Lap simulator as a useful learning tool that could challenge the basic skills of surgical trainees. In addition, all of the experienced participants rated the simulator as easy to use (Table 3), thus indicating that it could be an easy-to-use training device for surgical trainees.

The compact and easily transportable Ergo-Lap simulator provides opportunities for surgical trainees to practice their laparoscopic skills more conveniently, even at home or in the office. The integrated task panel for multiple training exercises can satisfy the need to practice both basic and advanced laparoscopic skills. The simulator also allows surgical trainees to develop their own exercises based on the panel. In addition, the computer software's recording capability can help trainees to preview previously recorded performance before starting

TABLE 3. Content Validity of the Ergo-Lap Simulator, as Assessed by the Experienced Group ($n = 20$), Rated Along a 5-Point Likert Scale (1 = Strongly Disagree [Very Poor], 3 = Neutral, and 5 = Strongly Agree [Excellent])

	Responses				
	1	2	3	4	5
How would you rate this Ergo-Lap simulator for training in an ergonomic manner? (e.g., ensuring proper manipulation angle/elevation angle/instrument scaling) *n* (%)	0	0	6 (30)	9 (45)	5 (25)
Do you consider this Ergo-Lap simulator useful for practicing basic laparoscopic skills? *n* (%)	0	0	0	11 (55)	9 (45)
Did you consider this Ergo-Lap simulator easy to use? *n* (%)	0	0	0	12 (60)	8 (40)
Would you recommend such an Ergo-Lap simulator to others? *n* (%)	1 (5)	0	3 (15)	10 (50)	6 (30)

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their training, in addition to helping them perform self-assessments and improve their skills after training. Based on the cost of the prototype, the anticipated price for the Ergo-Lap simulator is less than €300, which is within the expected price ranges stated by 46 participants (€200-€600). Compared with the expensive VR simulators, this Ergo-Lap simulator could offer an inexpensive alternative with which trainees can practice basic laparoscopic skills.

The present Ergo-Lap simulator nevertheless has several limitations that should be considered. First, the task materials do not provide an accurate reflection of the properties of real tissue. In addition, the simulator is not capable of identifying excessive force during the exercise involving the stretching of bands.

The issue of how to motivate surgical trainees to practice their laparoscopic skills voluntarily remains interesting.[29,30] Surgical trainees could use this webcam-based simulator to initiate an online laparoscopic skills competition amongst surgical trainees or experts throughout the world. This could also help to make surgical trainees more interested in using the simulator frequently, ultimately helping them to improve their skills. Future studies would involve this simulator in the Laparoscopic Surgical Skills (LSS) training program as a means of encouraging trainees to practice their basic skills before taking the formal course. For example, the Ergo-Lap simulator would be distributed to surgical trainees before the start of the LSS training program. Surgical trainees would not be entitled to register for the LSS training program until they have passed the basic skills test on the Ergo-Lap simulator. This would ensure that trainees could concentrate on advanced skills training using complicated arrangements in the skills laboratory during the LSS program.

CONCLUSIONS

Although simulation learning can never replace hands-on training, it does provide a safe and cost-effective environment within which surgeons can practice their laparoscopic skills. The novel Ergo-Laparoscopic simulator was found to be a useful, effective, and didactic training tool by both the Experienced and Novice participants in this study.

The 4 deliberately designed basic tasks were rated as good to excellent for training purposes. They were challenging, and they allowed the simulator to distinguish between different levels of expertise. Moreover, this Ergo-Lap simulator is an inexpensive alternative with which surgical trainees can update their skills in the skills laboratory, at home, or in the office.

DISCLOSURES

Authors D.J. Xiao, J.J. Jakimowicz, A. Albayrak, Sonja N. Buzink, Sanne M.B.I. Botden, and Richard Goossens have no conflicts of interest or financial ties to disclose.

ACKNOWLEDGMENTS

This study was supported by Catharina Hospital Eindhoven and Máxima Medical Centre, Veldhoven. We are very grateful to the participants involved in the study, as well as to Prof. A. Fingerhut for his valuable advice.

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